

PROJECT INTERIM REPORT

Movie Ticket Booking System

Developed using Core Java & OOP Principles

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Project Overview

Introduction

The **Movie Ticket Booking System** simulates a real-world cinema platform where users can book seats efficiently.

- Organized seat management using 2D arrays.
- Built on Object-Oriented Programming (OOP).
- Prevents errors like double-booking.
- Simple command-line interface for ease of use.

 Theater Seat Concept

System Working: Core Design



Modular Classes

Divided into **Movie** (details), **Seat** (pricing/type), and **Booking** (management) for better control.



Seat Layout

Utilizes a **2D Array** representing Rows and Columns to track every seat's booking status.



Price Logic

Automatically calculates total cost based on the number of seats selected and their category.



Status Flags

Uses simple status indicators:
'O' for Available seats
'X' for Reserved/Booked seats.

Implementation & Logic

```
// Seat Booking Logic

if (!seats[row][col].isBooked) {
    seats[row][col].book();
    total += price;
    System.out.println("Success!");
} else {
    System.out.println("Taken!");
}
```

Process Flow

- **Step 1:** System displays the current seating grid to the user.
- **Step 2:** User inputs their desired Row and Column index.
- **Step 3:** Program validates if coordinates are within range.
- **Step 4:** Checks isBooked boolean flag; if false, confirms booking.

Final Conclusion

Project Summary

The Movie Ticket Booking System successfully demonstrates the practical application of **Java OOP concepts**. It provides a scalable foundation for theater management.



Efficient Seat Tracking



Clean OOP Structure



Future Scalability

Image Sources



https://media.collegedekho.com/media/img/institute/logo/Screenshot_from_2025-08-19_12-01-11.png

Source: www.collegedekho.com



Thumbnail for
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